

Faithfulness from a unique irreducible class

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.injective_of_singleton`

theorem

Uses a pure state on the unitization to detect any hypothetical nonzero element of the kernel.

Statement

Let A be a nonzero complex C^* -algebra, with no unit assumed, and let $\pi : A \rightarrow \mathcal{B}(H)$ represent the unique unitary-equivalence class of nonzero irreducible $*$ -representations of A . Then π is injective:

$$\pi(x) = \pi(y) \implies x = y.$$

Assumptions

No separability of A or H is required. Faithfulness of the representation of the unitization is neither assumed nor concluded.

Conclusion

The original representation of A has zero kernel.

Proof route

A nonzero kernel element would be detected in a pure GNS representation of the unitization. Its nonzero irreducible restriction is compared with the given representation, forcing that detected operator to be zero.

▼ Proof steps

1. If $\pi(x) = \pi(y)$, put $a = x - y$, so $\pi(a) = 0$. Suppose $a \neq 0$ and let $\iota : A \rightarrow \tilde{A}$ be the canonical injective inclusion into the complex unitization. Then $\iota(a) \neq 0$. Apply Pure-state detection of a nonzero square in the nonzero unital ordered C^* -algebra $\tilde{A}$ to this element. It gives a pure state φ with $\varphi(\iota(a)^* \iota(a)) \neq 0$.
2. Let (H_φ, ρ_U, ξ) be its GNS representation with $\|\xi\| = 1$. By Irreducibility of the GNS representation of a pure state, purity makes ρ_U irreducible. Restrict it to A by $\rho(b) = \rho_U(\iota(b))$. The GNS identity gives

$$\|\rho(a)\xi\|^2 = \varphi(\iota(a)^* \iota(a)) \neq 0,$$

so $\rho(a) \neq 0$ and the restriction is a nonzero representation.

3. A closed subspace reducing $\rho(A)$ also reduces all $\rho_U(\lambda, b) = \lambda I + \rho(b)$, and conversely. Thus irreducibility of ρ_U , together with the nonzero restriction established in Step 2, gives nonzero irreducibility of ρ . This is the restriction criterion applied in the exact proof.

4. The singleton condition applied to this ρ supplies a unitary $U : H \rightarrow H_\varphi$ with $\rho(b)U = U\pi(b)$. Since U is surjective, every $z \in H_\varphi$ equals Uq for some $q \in H$, and

$$\rho(a)z = \rho(a)Uq = U\pi(a)q = 0.$$

Hence $\rho(a) = 0$, contradicting Step 2. Therefore $a = 0$ and $x = y$.

Main citations

- Exact theorem and proof
- Pure-state detection of a nonzero square
- Irreducibility of the GNS representation of a pure state
- The non-unital singleton predicate

Lean source signature (exact)

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theorem injective_of_singleton [Nontrivial A]
  (pi : NonUnitalCStarRepresentation A H)
  (hsingle : IsSingletonIrreducibleModel.{u, v, u} pi) :
  Function.Injective pi
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In the source	Mathematical meaning
[Nontrivial A]	The algebra is nonzero: $A \neq \{0\}$. This condition does not say whether a unit is assumed; that information comes from the surrounding C^* -algebra structure.
(pi : NonUnitalCStarRepresentation A H)	The specified $*$ -representation $\pi : A \rightarrow \mathcal{B}(H)$; no unit-preservation equation is required.
(hsingle : IsSingletonIrreducibleModel.{u, v, u} pi)	The specified representation is nonzero irreducible, and every nonzero irreducible $*$ -representation of the same algebra is unitarily equivalent to it.
Function.Injective pi	For all $x, y \in A$, equality of operators $\pi(x) = \pi(y)$ forces $x = y$. No separability binder occurs.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.injective_of_singleton

Accepted content SHA-256: 726046b9971256077466b6abf602fbeat45b711fab5f18643e07f1dc30fc8426

Accepted source guide SHA-256: 1ba4b9c85e4b62339a1564e27eddd8c98de64f97996aa740c4d07d3f91d20ce5

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73